

Week 6 Baseline Report

ESI Triage Classification

Mercer General Hospital · CariSurg MedTech Pathways

Dataset: yaleemmlc_admissionprediction_triage.csv — 55,121 patient visits, 226 columns

Target: esi (Emergency Severity Index, 1–5; 1 = most urgent, resuscitate)

Split: 80/20 stratified, random_state=42 (44,096 train / 11,025 test)

1. Dataset Recap

ESI is heavily imbalanced — this shapes every decision below:

ESI level	Count	Share
1 (Resuscitate)	77	0.1%
2 (Emergent)	17,924	32.5%
3 (Urgent)	27,010	49.0%
4 (Less urgent)	8,896	16.1%
5 (Non-urgent)	1,214	2.2%

ESI 1 is roughly 1 in 700 visits — only 16 land in the test set at all.

2. Models

- Logistic Regression — features scaled with StandardScaler (fit on train only).
- Decision Tree — max_depth=5, unscaled, chosen for interpretability over raw accuracy.
- Dummy (baseline) — strategy="stratified", the floor both real models must beat.

Features (208): 7 triage vitals, an O2-device flag, and 200 chief-complaint (cc_*) indicators.

Deliberately excluded: disposition/previousdispo (known only after triage — data leakage); demographics (age, gender, ethnicity, etc. — fairness-sensitive, held out of this baseline); administrative/arrival columns; and a leftover "Unnamed: 0" row-index column with no clinical meaning.

3. Results

Model	Accuracy	Macro F1	Recall (ESI 1)
Dummy (baseline)	0.375	0.204	0.00
Logistic Regression	0.667	0.492	0.25
Decision Tree	0.556	0.216	0.00

Both real models beat the dummy on accuracy and macro F1. Logistic Regression is the stronger baseline across the board, including on the metric that matters most clinically.

4. Metric Justification

Overall accuracy is misleading here — ESI 3 alone is half the data, so a model can score well while ignoring the rare, critical class entirely. Recall for ESI 1 is the metric we lead with: of the truly sickest patients, what fraction did the model catch? A missed ESI 1 is a delayed-care, possibly fatal error; a false alarm on a stable patient is costly but recoverable. Macro F1 is reported alongside weighted F1 (0.661) specifically because weighted F1 lets the large classes hide poor performance on the small ones — it looks reassuring in a way that isn't earned.

5. Failure Modes — Where It Falls Short

Recall on ESI 1 is 0.25 for Logistic Regression and 0.00 for the Decision Tree. Of 16 truly critical patients in the test set, Logistic Regression caught 4; the Decision Tree caught none.

Digging into why — comparing ESI 1 patients' vitals to the test set as a whole:

Vital	ESI 1 mean	All test patients
Heart rate	87.6	86.3
Systolic BP	152.6	133.9
Oxygen sat.	96.6	97.0
Resp. rate	21.0	17.8

Vitals for ESI 1 patients are not dramatically deranged on average — no collapsed blood pressure, no critically low oxygen. This is the "subtle presentation" problem: many of the sickest patients don't look obviously sick on the numbers alone.

The strongest available signal is the `cc_strokealert` chief-complaint flag, present in ~31% of ESI 1 patients. Reviewing exactly which patients each model missed: every single Logistic Regression catch was a stroke-alert case — it's the one flag strong enough to break through the class imbalance. The Decision Tree, at `max_depth=5`, never predicts ESI 1 at all; its learned splits favour the dominant classes. The remaining ESI 1 patients — presenting as dizziness, hypotension, mass, unresponsive, poisoning — share no common feature pattern, and each presentation is individually too rare for either model to learn from.

Clinical risk, stated plainly: as built today, this model would miss roughly 3 in 4 of the sickest patients, and is only reliably sensitive to one specific presentation (stroke alert). It cannot be used to deprioritize or wave through any patient without a clinician's independent judgment.

6. Does Hyperparameter Tuning Help? (Reviewer Follow-Up)

We tested `class_weight="balanced"`, which re-weights the training loss so rare classes count for more — no new data or features, just a different penalty for misclassifying them.

Model	Accuracy	Macro F1	Recall (ESI 1)
Logistic Regression (baseline)	0.667	0.492	0.250
Logistic Regression (balanced)	0.562	0.411	0.625
Decision Tree (baseline)	0.556	0.216	0.000
Decision Tree (balanced)	0.270	0.187	0.438

Balancing more than doubles Logistic Regression's ESI-1 recall (0.25 → 0.625). But it isn't a free win: overall accuracy drops (0.667 → 0.562) and macro F1 drops slightly too. Checking ESI-1 precision under the balanced model shows why — it falls to roughly 0.02, meaning the model now over-flags many more patients as ESI 1 to catch the true ones, trading a large increase in false alarms for a real gain in caught critical cases.

This is a policy decision, not a modeling one — how many false alarms is the ED willing to absorb per additional critical patient caught? That trade-off, plus proper resampling and threshold tuning, is what Week 7 will address directly.

7. Reproducibility

- `random_state=42` for the split and both models.
- Scaler fit on training data only, applied to test data — no leakage into scaling.
- Trained models saved: `model_logreg.joblib`, `model_tree.joblib`, `scaler.joblib`.
- Confusion matrices: `docs/w6_confusion_logreg.png`, `docs/w6_confusion_logreg_balanced.png`.